

# Resume

## Minwoo Lee

Bachelor Student, Kangnam University — Yongin, South Korea

Email: minwool0357@gmail.com | Phone: +82-10-6757-3689 | GitHub:

github.com/minuum

Portfolio: minuum-portfolio | Research Page: minuum.github.io/MoNaVLA

### EDUCATION

Kangnam University — Yongin, South Korea

Bachelor of Science in Artificial Intelligence (Senior, 1st Semester)

- Expected Graduation: Feb. 2027

- GPA: 4.27 / 4.5 (Ranked top of the department)

### RESEARCH INTERESTS

- Vision-Language-Action (VLA) Models for Robotics
- Embodied AI & Mobile Robot Navigation
- Sim-to-Real Transfer & Physical Latency Mitigation
- Deep Reinforcement Learning

### PUBLICATIONS & MANUSCRIPTS

- **EdgeGround-VLA: On-Device Goal-Directed Navigation via Dual Frozen Grounding Encoders**  
*Journal of The Korea Society of Computer and Information (JKSCI)*, KCI Registered, **Under Review (2026)**.  
**Minwoo Lee** (1st Author / Project Lead), et al.
  - Fuses dual frozen grounding encoders (OWLv2 + Kosmos-2) with a 0.866M-parameter action head (1.128M total trained parameters).
  - Achieves **95.0% real-robot goal-reaching success rate** fully on-device on Jetson Orin NX.
  - Project Page: minuum.github.io/EdgeGroundVLA
- **Designing Conversational AI Services Using RAG-Based Loneliness Analysis for Koreans**  
*Journal of the Korea Convergence Society (JCCT)*, KCI Registered, 2024.  
**Minwoo Lee** (2nd Author), et al.

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## CONFERENCES

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- Presented at the International Conference on IPACT 2024 —  
**Outstanding Paper Award**  
Minwoo Lee (2nd Author), et al.
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## RESEARCH EXPERIENCES

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**Undergraduate Research Assistant** — Advised by Prof. Inyeop Choe,  
Kangnam University

*Project: MoNaVLA (Mobile Navigation Vision-Language-Action Models) |  
Sep. 2025 – Present*

- **Diagnosis & Analysis:** Diagnosed structural text-attention collapse in the Kosmos-2 VLA backbone through per-layer attention measurement and frozen linear probe classification (val\_acc: 96.6%), revealing that post-training had silently destroyed the text pathway regardless of downstream fine-tuning. - **Insight Validation:** Confirmed that visual grounding drives navigation decisions via a target object masking ablation that induced 100% action reversal. - **Architecture Design:** Designed a two-stage decomposition pipeline integrating PaliGemma2 as a zero-shot visual grounder with the Kosmos-2 vision encoder, bounding box history, and L2 normalization. - **Performance:** Improved closed-loop navigation success rate from 10.3% (simple MLP baseline) to 66.7% (Decomposition v1) to **96.6%** (final, FPE: 0.094m) — a 9.4x gain from pipeline refinement alone on an identical grounding source, recovering from near-zero success in the end-to-end Kosmos-2 baseline. - **Real-Robot Validation:** Deployed the PaliGemma2 grounder on the physical Serbot 2 platform, raising real-robot grounding accuracy from 0% to 51.4% across 6 sessions (512 frames) by fixing camera-decoding bugs and applying bounding-box noise filtering; full closed-loop control testing is the next step. - **Award:** Won the **Silver Prize at the 2026-1 Kangnam University Capstone Design Competition.**

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## AWARDS AND HONORS

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- **Silver Prize**, 2026-1 Kangnam University Capstone Design Competition
- **Grand Prize (Dean of College of Engineering Award)**, 2025 Kangnam University Hackathon 'Kangnengthon' (Jointly hosted by Kangnam University & GDGoC: Kangnam University)
- **Outstanding Paper Award**, IPACT 2024 International Conference
- **First Prize (Best Project Award)**, Software Academic Festival (Kangnam University, 2024)
- **Excellence Award**, AI Academic Festival (Kangnam University, 2024)
- **Award**, K-HTML Hackathon (2024)

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## PROJECTS

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**Serbot 2 Robot Platform Integration** — Individual Project | *Jan. 2025 – Mar. 2025*

- Configured a Jetson Orin-based Linux environment and integrated ROS2 driver nodes for the Serbot 2 omnidirectional mobile robot platform. - Implemented core locomotion control scripts and configured the hardware-software communication interface to host VLA inference models. - Resolved physical-to-virtual command delays and stabilized actuator control signals.

**Ladi** — Software Academic Festival | *Oct. 2024*

- Developed an AI-based healthcare routine recommendation system (**Ladi**), winning the **First Prize**. - Implemented a hybrid search system combining dense (FAISS) and sparse (Kiwibm25) retrievers to optimize medical QA search. - Integrated a CrossEncoder Reranker to improve recommendation quality, and optimized prompt templates using LangSmith. - Preprocessed AIHub healthcare QA data to categorize diseases and route intents.

**Liberty** — AI Academic Festival | *May 2024*

- Served as Project Manager (PM) and lead architect for an AI Legal QA Agent system (**Liberty**), winning the **Excellence Award**. - Built a multi-agent workflow using LangGraph, integrating dynamic query rewriting and quick filtering. - Implemented a hybrid retriever (FAISS + KiwiBM25 + Pinecone) achieving 1.000 score in faithfulness and retrieval accuracy benchmarks.

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## SKILLS AND TECHNIQUES

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- **AI/ML:** PyTorch, HuggingFace, LoRA/PEFT, PaliGemma2, CLIP, SigLIP, Kosmos-2, LangGraph (Agent), LangSmith, FAISS, Pinecone (Vector DB), BM25, CrossEncoder (Reranker)
- **Robotics & Infra:** ROS2, Jetson Orin, Linux (Ubuntu), Locomotion Control, Asynchronous Control, Git
- **Programming Languages:** Python, C++, SQL
- **Languages:** Korean (Native), English (Intermediate — preparing for TOEIC)

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## COURSEWORK

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- **Deep Reinforcement Learning** — *Self-study (UC Berkeley CS285: Deep RL course lectures & homework)*
  - Focused on Imitation Learning, Policy Gradients, Actor-Critic methods, and value-based RL.